# Experiment 1

1.a. Analyze the trend of Data Science job postings over the last decade and visualize it using Pandas and Matplotlib.
1.b. Analyze and visualize the distribution of various Data Science roles from a dataset.
1.c. Differentiate Structured, Unstructured and Semi-structured data based on the given datasets.
1.d. Encrypt and decrypt sensitive data using the Cryptography library Fernet.

## Code 1

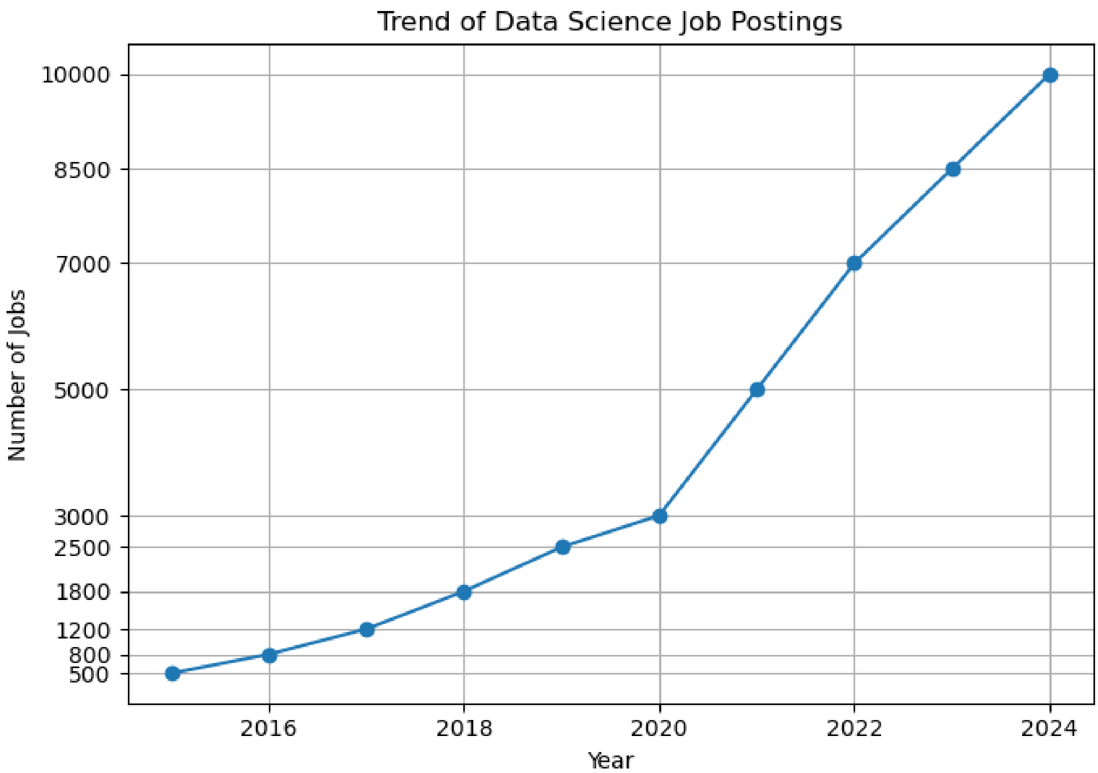
```
import pandas as pd
import matplotlib.pyplot as plt
years = [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]
jobs = [500, 800, 1200, 1800, 2500, 3000, 5000, 7000, 8500, 10000]
df_trend = pd.DataFrame({'Year': years, 'Number of Jobs': jobs})
print(df_trend)
plt.figure(figsize=(7, 5))
plt.plot(df_trend['Year'], df_trend['Number of Jobs'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Jobs')
plt.yticks(jobs)
plt.grid(True)
plt.tight_layout()
```

## Output

```
   Year  Number of Jobs
0  2015             500
1  2016             800
2  2017            1200
3  2018            1800
4  2019            2500
5  2020            3000
6  2021            5000
7  2022            7000
8  2023            8500
9  2024           10000
```

## Output

```
<Figure size 700x500 with 1 Axes>
```




## Code 2

```
import pandas as pd
import matplotlib.pyplot as plt
```

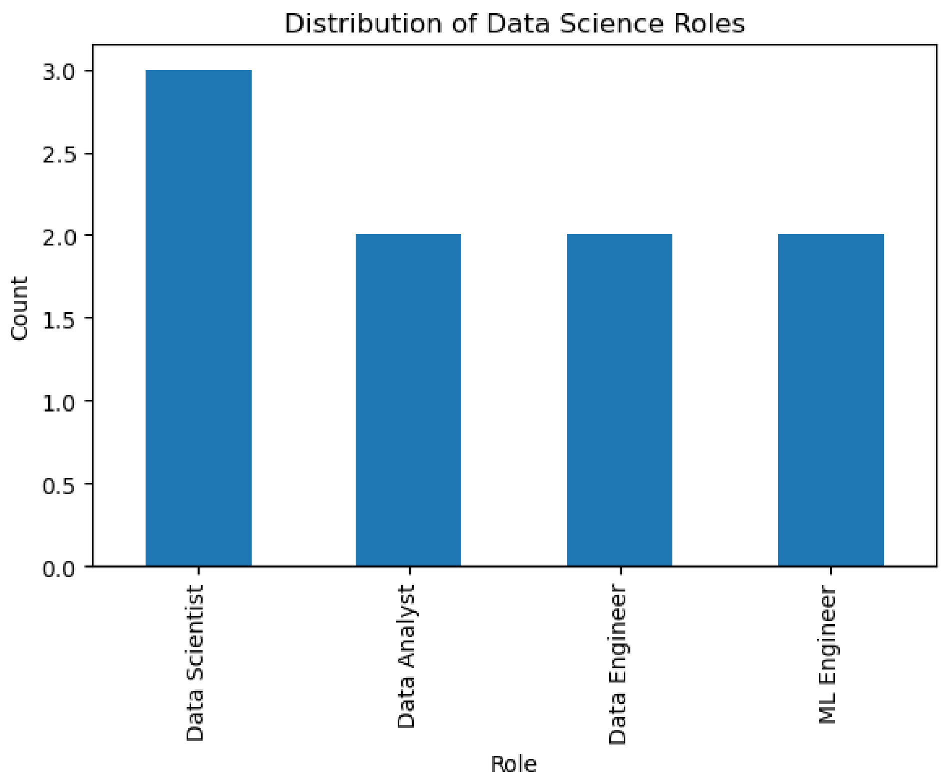
```
roles = ['Data Scientist', 'Data Scientist', 'Data Scientist',
 'Data Analyst', 'Data Analyst',
 'Data Engineer', 'Data Engineer',
 'ML Engineer', 'ML Engineer']
df_roles = pd.DataFrame({'Role': roles})
role_counts = df_roles['Role'].value_counts().reindex(
 ['Data Scientist', 'Data Analyst', 'Data Engineer', 'ML Engineer'])
print(role_counts)
plt.figure(figsize=(6, 5))
role_counts.plot(kind='bar')
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.xticks(rotation=90)
plt.tight_layout()
```

### Output

```
Role
Data Scientist    3
Data Analyst      2
Data Engineer     2
ML Engineer       2
Name: count, dtype: int64
```

### Output

```
<Figure size 600x500 with 1 Axes>
```




### Code 3

```
import pandas as pd
import json

structured_data = {
 'StudentID': [101, 102, 103],
 'Name': ['Arjun', 'Priya', 'Karthik'],
 'Marks': [88, 92, 79]
}
df_structured = pd.DataFrame(structured_data)
df_structured

semi_structured_data = '''
[
 {"StudentID": 101, "Name": "Arjun", "Marks": {"Math": 88, "Science": 91}},
 {"StudentID": 102, "Name": "Priya", "Subjects": ["Math", "English"], "Marks": 92}
]
'''
```

```
parsed = json.loads(semi_structured_data)
print(json.dumps(parsed, indent=2))

unstructured_data = """Student Arjun scored well in Math this term and showed
great improvement in Science."""
print(unstructured_data)
```

### Output

```
[
  {
    "StudentID": 101,
    "Name": "Arjun",
    "Marks": {
      "Math": 88,
      "Science": 91
    }
  },
  {
    "StudentID": 102,
    "Name": "Priya",
    "Subjects": [
      "Math",
      "English"
    ],
    "Marks": 92
  }
]
Student Arjun scored well in Math this term and showed
great improvement in Science.
```

### Code 4

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
cipher = Fernet(key)
print("Generated Key:", key.decode())

sensitive_data = "Student Name: Arjun, Aadhar No: 1234-5678-9012, Marks: 88"
print("\nOriginal Data:", sensitive_data)

data_bytes = sensitive_data.encode()
encrypted_data = cipher.encrypt(data_bytes)
print("\nEncrypted Data:", encrypted_data.decode())

decrypted_data = cipher.decrypt(encrypted_data)
print("\nDecrypted Data:", decrypted_data.decode())

print("\nMatch Original? ", decrypted_data.decode() == sensitive_data)
```

### Output

```
Generated Key: L48ZIAoAwJvTYIrJ3LvQY1qrMiWIFasVRUb8C1MfbGk=

Original Data: Student Name: Arjun, Aadhar No: 1234-5678-9012, Marks: 88

Encrypted Data: gAAAAABqdHuocUOs9xi3zmdx2vlzBQ2P2sYg3i6f5ut9hDuMr0v961lWUV05HmOgL8pcsMYuWV4bh54Gu3PiXRSPAvf8oQgEFJ63Hd

Decrypted Data: Student Name: Arjun, Aadhar No: 1234-5678-9012, Marks: 88

Match Original?  True
```